


# Assignment 01

**Submitted by:**
**Afifa Arooj**

**Registration No:**
**FA24B1-SE-003**

**Submitted To:**
**Waqas Saleem**

**Course:**
**Cloud Computing**

**Section:**
**BSSE-V A**

# Assignment 01: Advanced Git Operations, Dual-Remote Synchronization, Git LFS, and Submodules

## 1. Executive Summary

This report documents the step-by-step execution and completion of Assignment01. The lab assignment covers:

1. Local Git repository initialization and deployment on a self-hosted Gitea server.
2. Dual-remote configuration (origin for Gitea and github for GitHub) and cloud synchronization using Personal Access Tokens (PAT).
3. Installation and configuration of Git Large File Storage (Git LFS) to track and upload large binary files.
4. Integration of external Git repositories using Git Submodules.

## 2. Task 1: Gitea Deployment & Repository Initialization

2.1 Description

A self-hosted Gitea instance was deployed on the local Ubuntu VM. A new repository named Assignment01 was created on Gitea, and local commits including README.md and basic workflow directory structures were pushed to the origin remote.

### 2.2 Execution Steps

**Bash**

```
# Initialize git repository
git init

# Add local commit
git add README.md
git commit -m "Add student information"

# Add Gitea as origin remote and push
git remote add origin
http://192.168.184.129:3000/afifaarooj/Assignment01.git
git push -u origin main
```

### 2.3 Verification & Evidence

```
c5602f014e9e: Pull complete
ff5a46146cdc: Pull complete
83b8186c8e4a: Pull complete
c1bdaa07ad09: Pull complete
ddd2fa5fe1ae: Pull complete
d3a1c9de91ca: Pull complete
366619571da8: Download complete
64d311b93dec: Download complete
Digest: sha256:14bfab572eec6abf65892e1db7c3ba8d41b2a2855c1b143664907d6e146eb6e1
Status: Downloaded newer image for postgres:14
docker.io/library/postgres:14
afifaarooj@ubuntu:~/Gitea$ sudo docker pull gitea/gitea:1.21.7
1.21.7: Pulling from gitea/gitea
4b834617ab7f: Pull complete
619be1103602: Pull complete
d217270c7894: Pull complete
edc73e5c6bac: Pull complete
95f810368108: Pull complete
cccdca6bcafe: Pull complete
af8dc3e3859a: Pull complete
070ebb67d964: Download complete
Digest: sha256:daeeb33fb2f90d94876597f3b6c73956687c03250d5e5ca71104fc03b9c626b3
Status: Downloaded newer image for gitea/gitea:1.21.7
docker.io/gitea/gitea:1.21.7
afifaarooj@ubuntu:~/Gitea$ cd ~/Gitea && sudo docker-compose up -d && sudo docker-compose ps
[+] Running 1/7
 ✔ db 6 layers [      ]      0B/0B      Pulled                                                                                                  39.8s
   └ a8a481ae6efc Pulling fs layer                                                                                                             34.7s
   └ 7ccd0d0dcae74 Pulling fs layer                                                                                                            34.7s
   └ f3b07e8a357c Pulling fs layer                                                                                                             34.7s
   └ eb805f20f060 Pulling fs layer                                                                                                             34.7s
   └ 93a3470d5852 Pulling fs layer                                                                                                             34.7s
   └ b0a0d9d2abf2 Pulling fs layer                                                                                                             34.7s
[+] Running 2/5
 ⠿ Network webnet           Created                                                                                                            1.5s
 ⠿ Volume "gitea"           Created                                                                                                            1.3s
 ⠿ Volume "gitea_postgres"  Created                                                                                                            1.3s
 ✔ Container gitea_db       Started                                                                                                            1.3s
 ✔ Container gitea          Started                                                                                                            1.2s
NAME      IMAGE                COMMAND                  SERVICE   CREATED         STATUS                  PORTS
gitea     gitea/gitea:latest   "/usr/bin/entrypoint…"   gitea     2 seconds ago   Up Less than a second   22/tcp, 0.0.0.0:3000->3000/tcp, :::3000->3000/tcp
gitea_db  postgres:17-alpine   "docker-entrypoint.s…"   db        2 seconds ago   Up Less than a second   5432/tcp
afifaarooj@ubuntu:~/Gitea$
```

```
fatal: unable to auto-detect email address (got 'afifaarooj@ubuntu.(none)')
error: src refspec main does not match any
error: failed to push some refs to 'http://127.0.0.1:3000/afifaarooj/Assignment01.git'
afifaarooj@ubuntu:~/Assignment01$ ^[[200~git config --global user.name "afifaarooj"
git: command not found
afifaarooj@ubuntu:~/Assignment01$ git config --global user.email "afifaarooj@example.com"~
afifaarooj@ubuntu:~/Assignment01$ git config --global user.name "afifaarooj"
afifaarooj@ubuntu:~/Assignment01$ git config --global user.email "afifaarooj@example.com"
afifaarooj@ubuntu:~/Assignment01$ git commit -m "Add student information"
git push -u gitea main
[main (root-commit) 1d9bf8b] Add student information
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
Username for 'http://127.0.0.1:3000': afifaarooj
Password for 'http://afifaarooj@127.0.0.1:3000':
remote: Push to create is not enabled for users.
fatal: unable to access 'http://127.0.0.1:3000/afifaarooj/Assignment01.git/': The requested URL returned error: 403
afifaarooj@ubuntu:~/Assignment01$ git push -u gitea main
Username for 'http://127.0.0.1:3000': afifaarooj
Password for 'http://afifaarooj@127.0.0.1:3000':
remote: Push to create is not enabled for users.
fatal: unable to access 'http://127.0.0.1:3000/afifaarooj/Assignment01.git/': The requested URL returned error: 403
afifaarooj@ubuntu:~/Assignment01$ git remote set-url gitea http://192.168.184.129:3000/afifaarooj/Assignment01.git
git push -u gitea main
Username for 'http://192.168.184.129:3000': afifaarooj
Password for 'http://afifaarooj@192.168.184.129:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 233 bytes | 116.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://192.168.184.129:3000/afifaarooj/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
afifaarooj@ubuntu:~/Assignment01$
```

Task1_gitea_running.png

```
          uname -a
          docker --version
EOF
afifaarooj@ubuntu:~/Assignment01$ git add .gitea/workflows/test.yml
git commit -m "Add workflow"
git push origin main
[main 6cfcb3b] Add workflow
 1 file changed, 15 insertions(+)
 create mode 100644 .gitea/workflows/test.yml
fatal: 'origin' does not appear to be a git repository
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.
afifaarooj@ubuntu:~/Assignment01$ git remote add origin http://192.168.184.129:3000/afifaarooj/Assignment01.git
afifaarooj@ubuntu:~/Assignment01$ git branch -M main
afifaarooj@ubuntu:~/Assignment01$ git push -u origin main
Username for 'http://192.168.184.129:3000': afifaarooj
Password for 'http://afifaarooj@192.168.184.129:3000':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 557 bytes | 278.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://192.168.184.129:3000/afifaarooj/Assignment01.git
   1d9bf8b..6cfcb3b  main -> main
branch 'main' set up to track 'origin/main'.
afifaarooj@ubuntu:~/Assignment01$
```

task1_gitea_push.png

task1_gitea_repository.png

# 3. Task 2: Dual Remote Setup & GitHub Synchronization

### 3.1 Description

To achieve dual-remote synchronization, a secondary remote named github was linked to the local repository. Authentication was successfully executed using a Personal Access Token (PAT) due to GitHub's password authentication restrictions.

### 3.2 Execution Steps

Bash

```
# Configure secondary GitHub remote

git remote add github https://github.com/afifaarooj/Assignment01.git



# Verify configured remotes

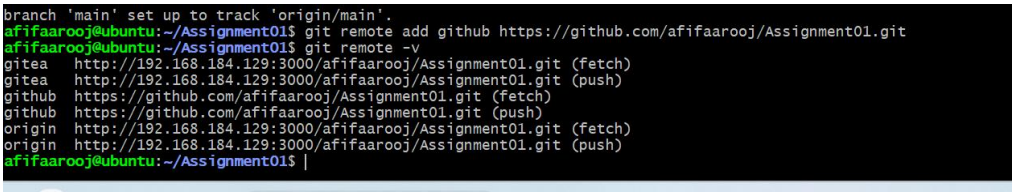
git remote -v



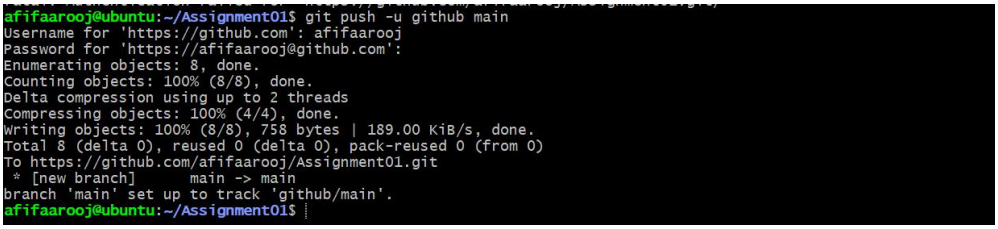
# Push main branch to GitHub using PAT authentication

git push -u github main
```

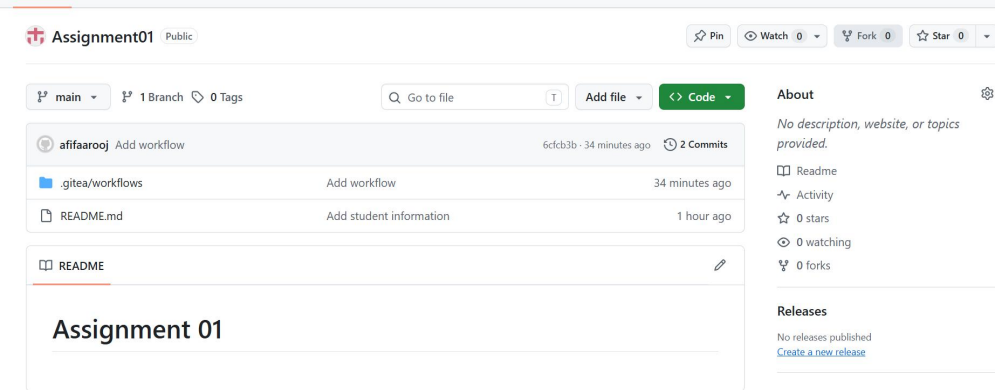
## 3.3 Verification & Evidence




task2_remotes.png




task2_github_push.png




Task2_github_repository.png

# 4. Task 3: Track Three Large Files with Git LFS

## 4.1 Description

Standard Git repositories struggle with large binary files. Git LFS was installed and configured to track .bin binary files. Three dummy files (10MB each) were generated using the dd command, tracked via .gitattributes, committed, and pushed to the remote repository.

## 4.2 Execution Steps

### Bash

```
# Install and initialize Git LFS
sudo apt update && sudo apt install git-lfs
git lfs install

# Generate three 10MB dummy files
dd if=/dev/zero of=file1.bin bs=1M count=10
dd if=/dev/zero of=file2.bin bs=1M count=10
dd if=/dev/zero of=file3.bin bs=1M count=10

# Configure LFS tracking pattern and stage attributes
git lfs track "*.bin"
git add .gitattributes

# Stage, commit, and push LFS tracked files
git add file1.bin file2.bin file3.bin
git commit -m "Add three large files tracked with Git LFS"
git push github main
```

## 4.3 Verification & Evidence

```
branch 'main' set up to track 'github/main'.
afifaarooj@ubuntu:~/Assignment01$ sudo apt update
sudo apt install git-lfs
git lfs install
[sudo: authenticate] Password:
Hit:1 http://pk.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://pk.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://pk.archive.ubuntu.com/ubuntu resolute-backports InRelease
56 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
  linux-tools-7.0.0-30  linux-tools-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  git-lfs

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 56
  Download size: 4,309 kB
  Space needed: 13.1 MB / 309 MB available

Get:1 http://pk.archive.ubuntu.com/ubuntu resolute/universe amd64 git-lfs amd64 3.7.1-1 [4,309 kB]
Fetched 4,309 kB in 9s (476 kB/s)
Selecting previously unselected package git-lfs.
(Reading database… 99082 files and directories currently installed.)
Preparing to unpack …/git-lfs_3.7.1-1_amd64.deb…
Unpacking git-lfs (3.7.1-1)…
Setting up git-lfs (3.7.1-1)…
Processing triggers for man-db (2.13.1-1build1)…
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
Updated Git hooks.
Git LFS initialized.
afifaarooj@ubuntu:~/Assignment01$ |
```

task3_lfs_install.png

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
Updated Git hooks.
Git LFS initialized.
afifaarooj@ubuntu:~/Assignment01$ dd if=/dev/zero of=file1.bin bs=1M count=10
10+0 records in
10+0 records out
10485760 bytes (10 MB, 10 MiB) copied, 0.0131116 s, 807 MB/s
afifaarooj@ubuntu:~/Assignment01$ dd if=/dev/zero of=file2.bin bs=1M count=10
10+0 records in
10+0 records out
10485760 bytes (10 MB, 10 MiB) copied, 0.0117288 s, 953 MB/s
afifaarooj@ubuntu:~/Assignment01$ dd if=/dev/zero of=file3.bin bs=1M count=10
10+0 records in
10+0 records out
10485760 bytes (10 MB, 10 MiB) copied, 0.0129126 s, 874 MB/s
afifaarooj@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
afifaarooj@ubuntu:~/Assignment01$ git add .gitattributes
afifaarooj@ubuntu:~/Assignment01$ git status
On branch main
Your branch is up to date with 'github/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   .gitattributes

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file1.bin
        file2.bin
        file3.bin

afifaarooj@ubuntu:~/Assignment01$ |
```

task3_lfs_track.png

```
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file1.bin
        file2.bin
        file3.bin

afifaarooj@ubuntu:~/Assignment01$ git add file1.bin file2.bin file3.bin
afifaarooj@ubuntu:~/Assignment01$ git commit -m "Add three large files tracked with Git LFS"
[main fea3019] Add three large files tracked with Git LFS
 4 files changed, 10 insertions(+)
 create mode 100644 .gitattributes
 create mode 100644 file1.bin
 create mode 100644 file2.bin
 create mode 100644 file3.bin
afifaarooj@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': afifaarooj
Password for 'https://afifaarooj@github.com':
Username for 'https://github.com': afifaarooj
Password for 'https://afifaarooj@github.com':
Enumerating objects: 5, done.1/1), 10 MB | 230 KB/s
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 537 bytes | 537.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/afifaarooj/Assignment01.git
   6cfcb3b..fea3019  main -> main
afifaarooj@ubuntu:~/Assignment01$ |
```

task3_lfs_push.png

# 5. Task 4: Git Submodule Integration

## 5.1 Description

To incorporate external dependencies cleanly without direct code copying, a public GitHub repository (octocat/Hello-World) was added as a Git Submodule inside the my-submodule directory. The submodule metadata (.gitmodules) was committed and synchronized.

### 5.2 Execution Steps

Bash
# Add public repository as a submodule
git submodule add https://github.com/octocat/Hello-world.git my-submodule

# Verify staging status
git status

# Commit submodule additions
git commit -m "Add Hello-World repository as a submodule"

# Push updates to GitHub remote
git push github main

### 5.3 Verification & Evidence




task4_submodule_add.png




task4_submodule_push.png

# 6. Conclusion

All four core objectives specified in **Assignment01** were successfully accomplished. The local repository is fully functional with dual remotes (Gitea and GitHub), binary tracking via Git LFS, and modular external linkage using Git Submodules.